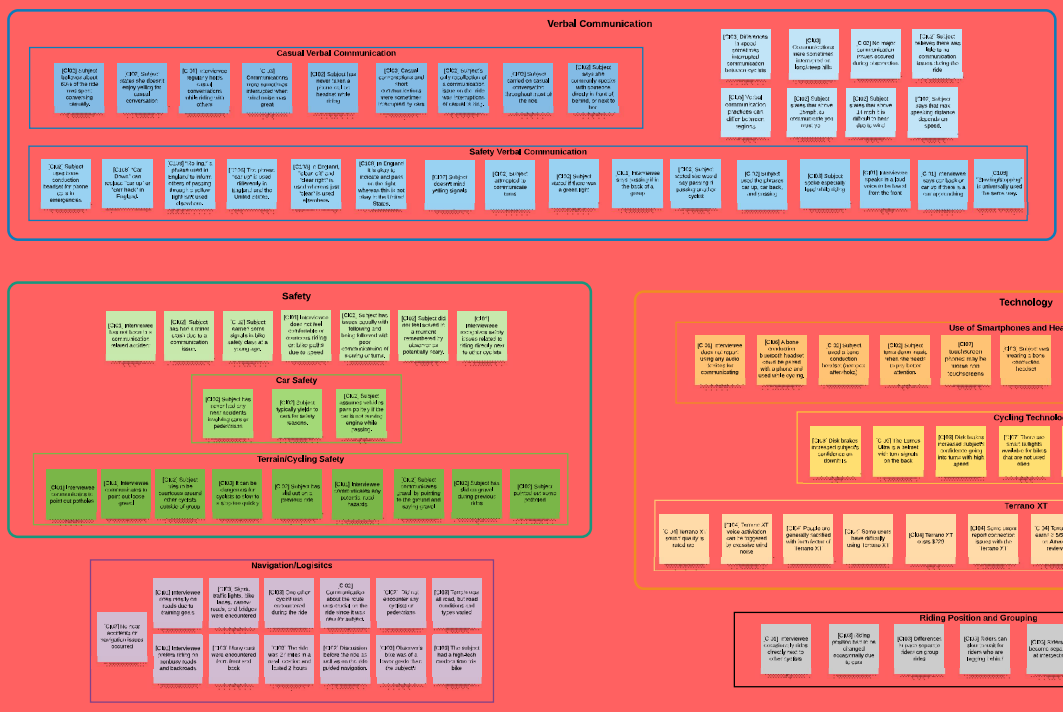
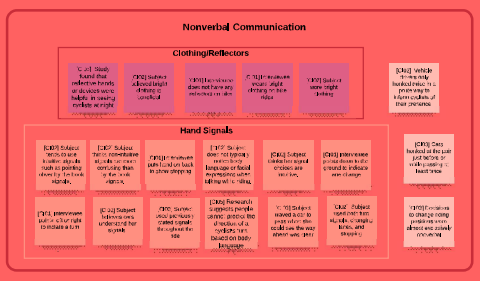
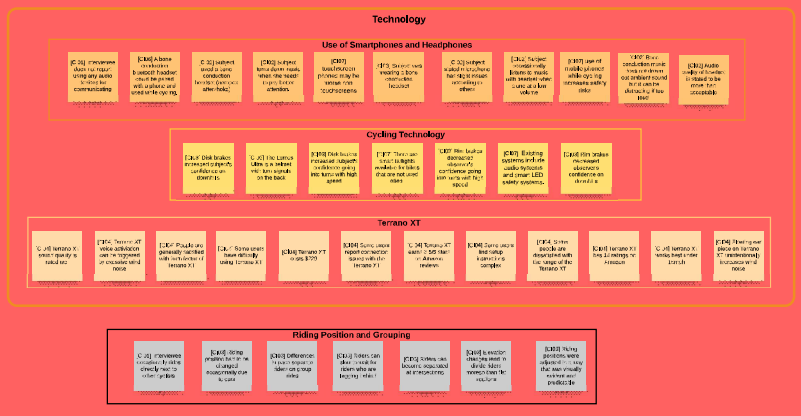
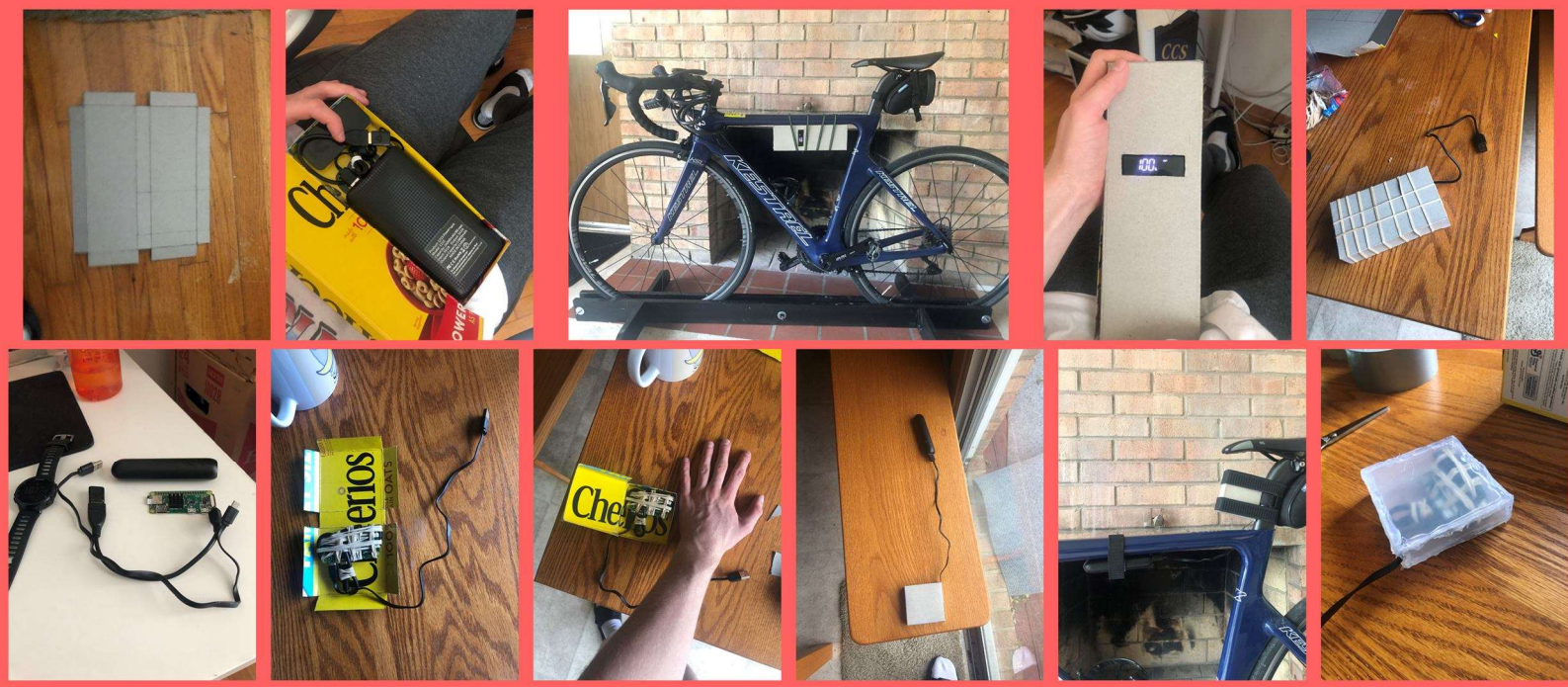
BIKE BROADCAST

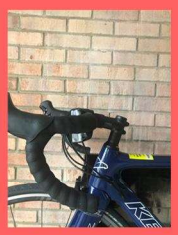
Bike Broadcast was a semester-long undergraduate research project. I led two other students with whom I collaborated on the Work Activity Affinity Diagram directly below. The app shown is an open-source application. The remainder of the images are my own original designs and analysis. My final prototype allowed cyclists to talk wirelessly in real-time at a range of up to 500 feet without the need for cellular service, using iPhones and the headset shown.

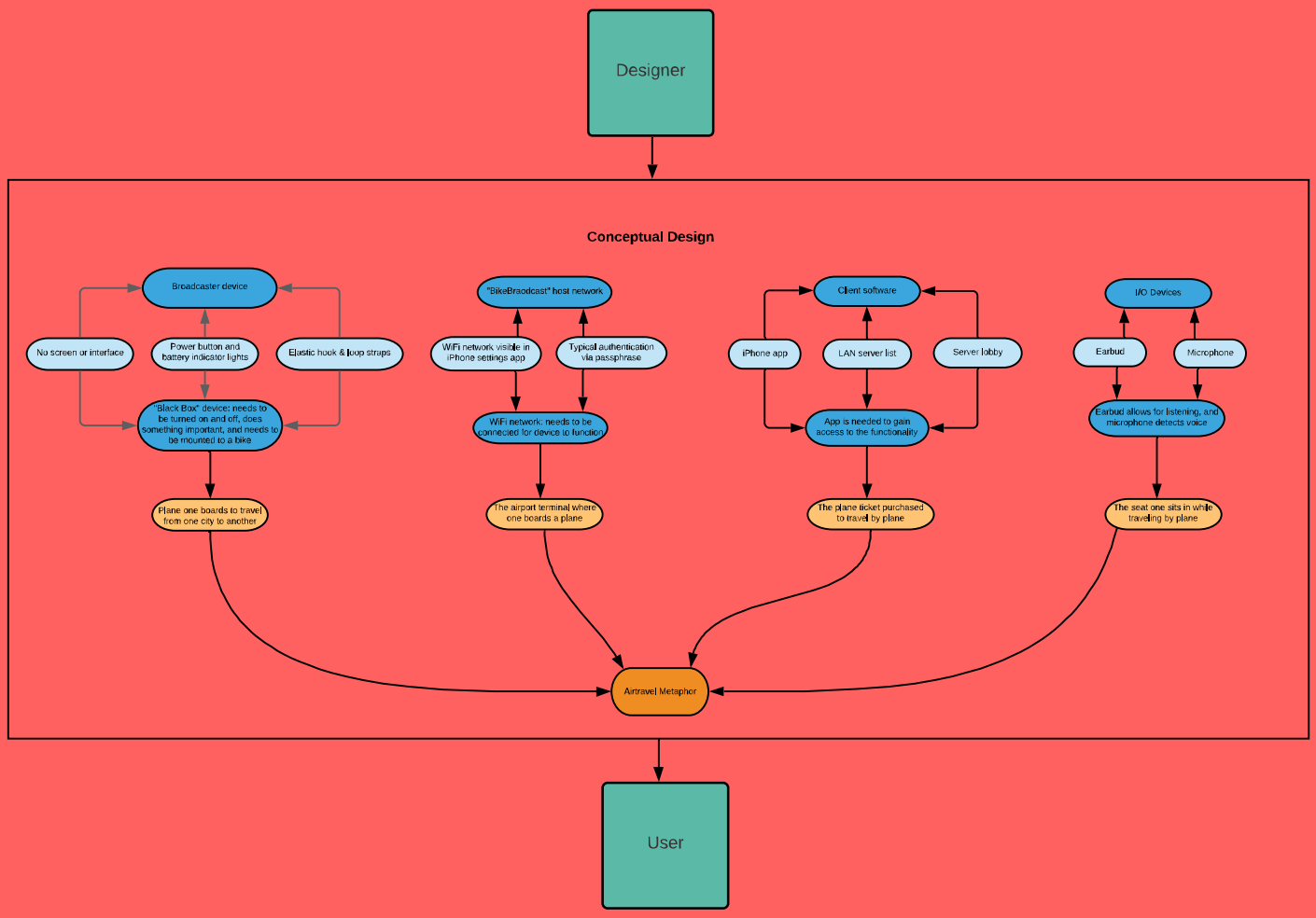


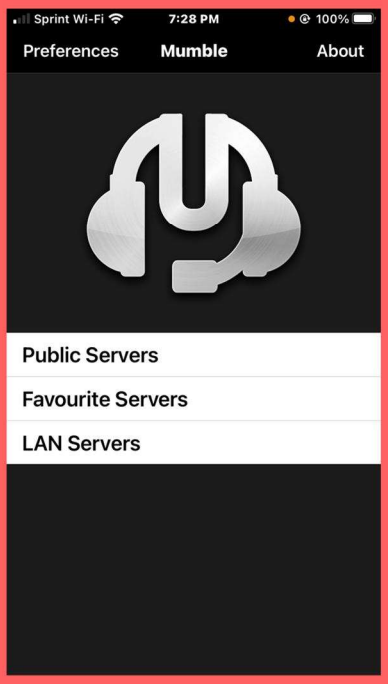









Designer
Conceptual Design
Broadcaster device
No screen or interface
Power button and battery indicator lights
Elastic hook & loop straps
"Black Box" device: needs to be turned on and off, does something important, and needs to be mounted to a bike
Plane one boards to travel from one city to another
"BikeBraodcast" host network
WiFi network visible in iPhone settings app
Typical authentication via passphrase
WiFi network: needs to be connected for device to function
The airport terminal where one boards a plane
Client software
iPhone app
LAN server list
Server lobby
App is needed to gain access to the functionality
The plane ticket purchased to travel by plane
I/O Devices
Earbud
Microphone
Earbud allows for listening, and microphone detects voice
The seat one sits in while traveling by plane
Airtravel Metaphor
User


Sprint Wi-Fi
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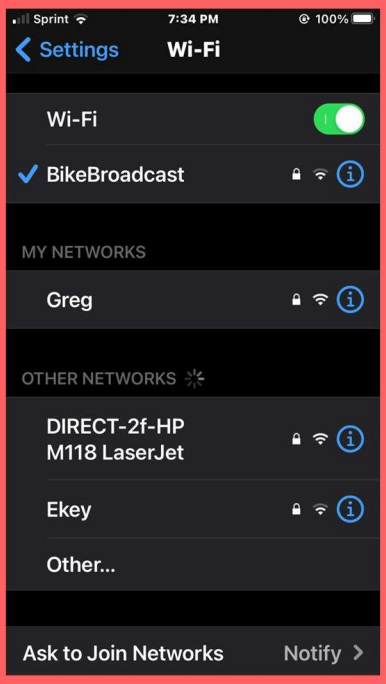
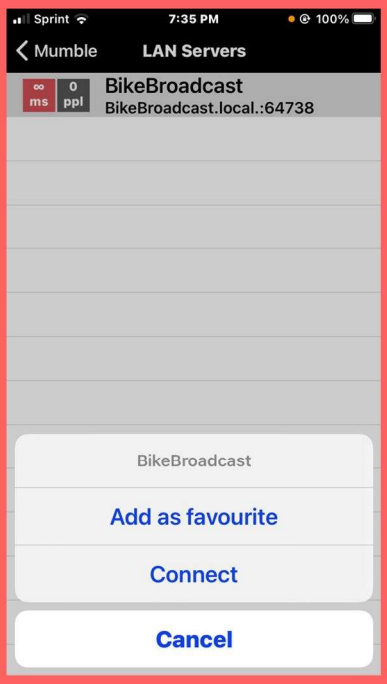
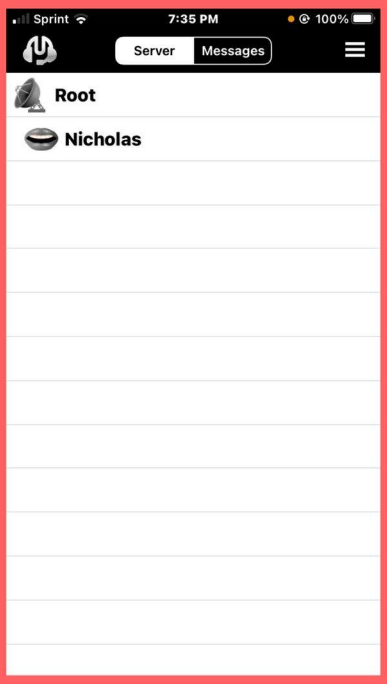
100%
Preferences
Mumble
About
Public Servers
Favourite Servers
LAN Servers


Sprint
7:34 PM
100%
Settings
Wi-Fi
Wi-Fi
BikeBroadcast
MY NETWORKS
Greg
OTHER NETWORKS
DIRECT-2f-HP
M118 LaserJet
Ekey
Other...
Ask to Join Networks
Notify


Sprint
7:35 PM
100%
Mumble
LAN Servers
∞ ms
0 ppl
BikeBroadcast
BikeBroadcast.local.:64738
BikeBroadcast
Add as favourite
Connect
Cancel


Sprint
7:35 PM
100%
Server
Messages
Root
Nicholas